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Traction image method for irregular free surface boundaries in finite difference seismic wave simulation

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SUMMARY

In this study, we propose a new numerical method, named as Traction Image method, to accurately and efficiently implement the traction-free boundary conditions in finite difference simulation in the presence of surface topography. In this algorithm, the computational domain is discretized by boundary-conforming grids, in which the irregular surface is transformed into a 'flat' surface in computational space. Thus, the artefact of staircase approximation to arbitrarily irregular surface can be avoided. Such boundary-conforming gridding is equivalent to a curvilinear coordinate system, in which the first-order partial differential velocity-stress equations are numerically updated by an optimized high-order non-staggered finite difference scheme, that is, DRP/opt MacCormack scheme. To satisfy the free surface boundary conditions, we extend the Stress Image method for planar surface to Traction Image method for arbitrarily irregular surface by antisymmetrically setting the values of normal traction on the grid points above the free surface. This Traction Image method can be efficiently implemented. To validate this new method, we perform numerical tests to several complex models by comparing our results with those computed by other independent accurate methods. Although some of the testing examples have extremely sloped topography, all tested results show an excellent agreement between our results and those from the reference solutions, confirming the validity of our method for modelling seismic waves in the heterogeneous media with arbitrary shape topography. Numerical tests also demonstrate the efficiency of this method. We find about 10 grid points per shortest wavelength is enough to maintain the global accuracy of the simulation. Although the current study is for 2-D *P-SV* problem, it can be easily extended to 3-D problem.

Key words: finite difference, MacCormack scheme, seismic wave simulation, stress image method, topography, traction Image method.

1 INTRODUCTION

Finite difference method (FDM) is one of the most popular numerical methods used in seismological study. During past three decades, FDM had been widely used in both the modelling of rupture dynamics of earthquake source (Madariaga 1976; Andrews 1976a,b; Olsen *et al.* 1997; Madariaga *et al.* 1998; Cruz-Atienza & Virieux 2004), and the simulation of seismic wave propagation in complex heterogeneous media (Boore 1972; Kelly *et al.* 1976; Bayliss *et al.* 1986; Virieux 1984, 1986; Levander 1988; Graves 1996; Dai *et al.* 1995; Zahradnik 1995). The majority of these studies used interior complex geological models but with a planar surface, because there is still lack of a commonly acceptable numerical free surface treatment for FDM in the presence of surface topography (Jih *et al.* 1988; Frankel & Leith 1992; Oprsal & Zahradnik 1999; Ohminato & Chouet 1997; Robertsson 1996; Pitarka & Irikura 1996; Hestholm & Ruud 1994, 1998).

How to accurately satisfy the traction-free boundary conditions on a general topographic surface is an important and difficult issue for finite difference simulation based on the first-order partial differential velocity-stress equations. Even for a planar surface, it is not yet a trivial problem (Gottschammer & Olsen 2001; Kristek *et al.* 2002). For planar surface, the traction-free boundary condition can be implemented by the Vacuum method (Boore 1972; Graves 1996), the Characteristic Variables method (Bayliss *et al.* 1986), AFDA (adjusted FD approximations) technique (Kristek *et al.* 2002), and the Stress Image method (Levander 1988; Graves 1996).

Several efforts extending some of the above successful methods to the problem with general irregular topography have been made. With staircase approximation to irregular surfaces, Ohminato & Chouet (1997) extended the Stress Image method to the general topographic problem in the second-order accurate staggered finite difference scheme. Although the authors claimed their method was a vacuum method, they increased the computed normal derivatives of stress components by a factor of 2 at the free surface, which is equivalent to the implementation of Stress Image method. Though the Stress Image method can be precisely implemented in second-order staggered scheme, the results of simulation still surfers from the

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discretization error due to the staircase approximation to the surface topography. Furthermore, the second-order schemes contain significant numerical dispersion error. According to Ohminato & Chouet (1997), 20 grid points per shortest wavelength are needed to suppress such numerical dispersion error even for the case in which the wavefields propagate only several wavelengths away from the source. For wave propagation over longer distances, the grid resolution requirements of the second-order staggered scheme will become excessive. Robertsson (1996) and Pitarka & Irikura (1996), independently, implemented the Stress Image method in the fourth-order staggered scheme by using the staircase approximation to the irregular surface. Compared with the precise treatment of the stress components (Robertsson 1996), the treatment of the velocity components at the free surface in this fourth-order scheme is approximate, because the velocity components above the free surface were simply set to vanish. Moreover, the staircase approximation can generate artificial scattering waves (Pitarka & Irikura 1996), which may affect the physical scattering waves or multiple reflection waves.

To avoid the staircase approximation, Hestholm & Ruud (1994, 1998) used a vertical grid mapping to match the computational grids with the surface topography in staggered finite difference schemes. Because the stress-imaging technique can not be applied on such non-Cartesian grids, they only used the velocity constraint derived from the free surface boundary conditions, consequently, the Rayleigh waves can not be properly simulated.

In this study, we shall develop a new algorithm to tackle the topography problem, the long standing difficult issue in finite difference simulation. To begin with, we briefly describe boundary-conforming grid to discretize the computational domain. Then, the first-order partial differential velocity-stress equations are expanded in curvilinear coordinates. Furthermore, a non-staggered finite difference scheme, DRP/opt MacCormack scheme, is used to update the these equations. Finally, the Traction Image method to calculate the stress derivatives and a compact scheme to calculate the velocity derivatives on the grid points near the free surface are described in detail, along with the numerical validations and discussions.

2 BOUNDARY-CONFORMING GRID

When surface topography presents, the discrete grid must conform to the free surface to avoid artificial scattering waves. Such grid is named as boundary-conforming grid (Fig. 1), which was used in seismic wave simulation by Fornberg (1988) with pseudospectral method to make the computational grid align with media interfaces. The boundary-conforming grid transforms the irregular surface in the physical space $x = x(x, y)$ into a 'flat' surface in the computational space $X = X(\xi, \eta)$, as shown in Fig. 1. Then the free surface conditions can be implemented on such a 'flat' surface.

Once the free surface topography is given, algebraic or PDE (partial differential equations) methods can be used to generate the boundary-conforming grid. The grid qualities, estimated in terms of concentration of grid points, smoothness of the variations in the grid size and orthogonality of the grid, can be controlled during generation (Thompson *et al.* 1985). Numerical grid generation as a basic step in many scientific computing problems has been developed over a long period of time and has become a fairly common tool in the numerical solution of partial differential equations (Thompson *et al.* 1985). Lots of commercial and free software are available (In this study, the grid is generated by using the ICEM CDF software of ANSYS Inc.).

After the boundary-conforming grid being generated, the Cartesian coordinate of each discrete grid points can be determined. Thus the mapping from the curvilinear coordinate to the Cartesian coordinate is,

$$\begin{aligned} x &= x(\xi, \eta) \\ y &= y(\xi, \eta). \end{aligned} \tag{1}$$

And their derivatives can be numerically calculated,

$$\begin{aligned} x_{,\xi} &= D_\xi x, \\ x_{,\eta} &= D_\eta x, \\ y_{,\xi} &= D_\xi y, \\ y_{,\eta} &= D_\eta y, \end{aligned} \tag{2}$$

where, $x_{,\xi}$, $x_{,\eta}$, $y_{,\xi}$, and $y_{,\eta}$ are shorthand representations of the derivatives $\partial x/\partial\xi$, $\partial x/\partial\eta$, $\partial y/\partial\xi$, $\partial y/\partial\eta$. For simplicity, we use such representations throughout this article. D_ξ and D_η represent the difference operators of the spatial derivatives ∂_ξ and ∂_η. It is worth to note that even if the mapping eq. (1) are given by an analytical function, the transforming derivatives should still be calculated numerically to avoid spurious source terms caused by the coefficients derivatives when conservation form momentum equations are used (Thompson *et al.* 1985).

From the relations

$$\begin{aligned} x_{,\xi}\,\xi_{,x} + x_{,\eta}\,\eta_{,x} &= 1, \\ x_{,\xi}\,\xi_{,y} + x_{,\eta}\,\eta_{,y} &= 0, \\ y_{,\xi}\,\xi_{,y} + y_{,\eta}\,\eta_{,y} &= 1, \\ y_{,\xi}\,\xi_{,x} + y_{,\eta}\,\eta_{,x} &= 0, \end{aligned} \tag{3}$$

we can derived,

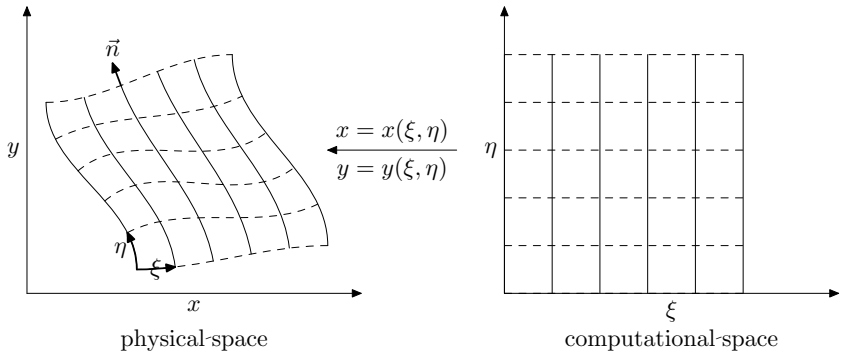


Figure 1. Mapping between the boundary-conforming grid in the physical space $x = x(x, y)$ and the uniform grid in the computational space X = X (ξ, η).

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$$\xi_{,x} = \frac{1}{J}\, y_{,\eta}, \qquad \xi_{,y} = -\frac{1}{J}\, x_{,\eta},$$
$$\eta_{,x} = -\frac{1}{J}\, y_{,\xi}, \qquad \eta_{,y} = \frac{1}{J}\, x_{,\xi},$$
$$J = (x_{,\xi} y_{,\eta} - x_{,\eta} y_{,\xi}). \tag{4}$$

In these equations, J is the Jacobian of the transformation and is a non-zero value (Thompson *et al.* 1985).

Choosing computational grid aligned with the free surface and setting the ξ axis parallel to surface, we can derive the unit vector perpendicular to the free surface as follows (see Appendix A)

$$\vec{n} = \frac{\eta_{,x}\vec{i}_1 + \eta_{,y}\vec{i}_2}{|\nabla\eta|}, \tag{5}$$

where $\vec{i}_1$ and $\vec{i}_2$ are the unit vectors of the x axis and the y axis of the Cartesian coordinate, and $|\nabla\eta| = \sqrt{\eta_{,x}^2 + \eta_{,y}^2}$.

3 ELASTIC WAVE EQUATIONS IN GENERAL CURVILINEAR COORDINATES

For 2-D inhomogeneous linear isotropic elastic media, the first-order partial differential velocity-stress equations are written as:

$$\rho v_{x,t} = \sigma_{xx,x} + \sigma_{xy,y} + f_x, \tag{6a}$$

$$\rho v_{y,t} = \sigma_{xy,x} + \sigma_{yy,y} + f_y, \tag{6b}$$

$$\sigma_{xx,t} = (\lambda + 2\mu) v_{x,x} + \lambda v_{y,y}, \tag{6c}$$

$$\sigma_{yy,t} = \lambda v_{x,x} + (\lambda + 2\mu) v_{y,y}, \tag{6d}$$

$$\sigma_{xy,t} = \mu(v_{x,y} + v_{y,x}), \tag{6e}$$

where, v_x and v_y are, respectively, the x- and y-components of particle velocity, σ_{xx}, σ_{yy}, and σ_{xy} are the Cartesian components of the stress tensor, f_x and f_y are body forces, ρ is density, λ and μ are Lame parameters.

The velocity-stress eqs (6a and b) must be transformed into curvilinear coordinates when the boundary-conforming grid is applied. In curvilinear coordinates, the above equations become

$$\rho v_{x,t} = \xi_{,x}\sigma_{xx,\xi} + \xi_{,y}\sigma_{xy,\xi} + \eta_{,x}\sigma_{xx,\eta} + \eta_{,y}\sigma_{xy,\eta}, \tag{7a}$$

$$\rho v_{y,t} = \xi_{,x}\sigma_{xy,\xi} + \xi_{,y}\sigma_{yy,\xi} + \eta_{,x}\sigma_{xy,\eta} + \eta_{,y}\sigma_{yy,\eta}, \tag{7b}$$

$$\sigma_{xx,t} = (\lambda + 2\mu)\xi_{,x} v_{x,\xi} + \lambda\xi_{,y} v_{y,\xi} + (\lambda + 2\mu)\eta_{,x} v_{x,\eta} + \lambda\eta_{,y} v_{y,\eta}, \tag{7c}$$

$$\sigma_{yy,t} = \lambda\xi_{,x} v_{x,\xi} + (\lambda + 2\mu)\xi_{,y} v_{y,\xi} + \lambda\eta_{,x} v_{x,\eta} + (\lambda + 2\mu)\eta_{,y} v_{y,\eta}, \tag{7d}$$

$$\sigma_{xy,t} = \mu\left(\xi_{,y} v_{x,\xi} + \xi_{,x} v_{y,\xi} + \eta_{,y} v_{x,\eta} + \eta_{,x} v_{y,\eta}\right). \tag{7e}$$

In these equations (eqs 7a and b) represent the momentum equation, and eqs (7c–e) are the generalized Hook's Law. The source terms are omitted here and will be discussed in Section 5.

The above equations can be rewritten in a matrix form

$$W_{,t} = A W_{,\xi} + B W_{,\eta}, \tag{8}$$

where

$$W = [v_x, v_y, \sigma_{xx}, \sigma_{yy}, \sigma_{xy}]^T,$$

$$A = \begin{bmatrix} 0 & 0 & \xi_{,x}/\rho & 0 & \xi_{,y}/\rho \\ 0 & 0 & 0 & \xi_{,y}/\rho & \xi_{,x}/\rho \\ \chi\xi_{,x} & \lambda\xi_{,y} & 0 & 0 & 0 \\ \lambda\xi_{,x} & \chi\xi_{,y} & 0 & 0 & 0 \\ \mu\xi_{,y} & \mu\xi_{,x} & 0 & 0 & 0 \end{bmatrix},$$

$$B = \begin{bmatrix} 0 & 0 & \eta_{,x}/\rho & 0 & \eta_{,y}/\rho \\ 0 & 0 & 0 & \eta_{,y}/\rho & \eta_{,x}/\rho \\ \chi\eta_{,x} & \lambda\eta_{,y} & 0 & 0 & 0 \\ \lambda\eta_{,x} & \chi\eta_{,y} & 0 & 0 & 0 \\ \mu\eta_{,y} & \mu\eta_{,x} & 0 & 0 & 0 \end{bmatrix},$$

which is convenient for the implementation of non-staggered finite difference schemes. In the above equations, $\chi = \lambda + 2\mu$.

4 DRP/OPT MACCORMACK SCHEME

In this section, we describe the finite difference scheme to update eqs (7a–e), or equivalently eq. (8), while omitting the boundary treatment, which will be discussed in Section 5.

Although ideal on Cartesian grids, staggered-grid finite difference schemes are not well suited on non-Cartesian grids because complex interpolations are needed and decrease the overall accuracy (Magnier *et al.* 1994; Igel *et al.* 1995; Komatitsch *et al.* 1996). The implementation of non-staggered grid schemes is more preferable on boundary-conforming grids. One popular and well-tested non-staggered scheme is MacCormack-type scheme. MacCormack-type scheme has inherent dissipation, which damps the spurious short-wavelength numerical (or non-physical) waves generated by media discontinuities, computational domain boundaries, grid discontinuities and other computational irregularities, and has a negligible effect on the physical solution.

The original MacCormack scheme (MacCormack 1969) is second-order accurate in both time and space, and was extended to second-order accurate in time and fourth-order accurate in space by Gottlieb & Turkel (1976). This extension of the MacCormack scheme, known as 2–4 MacCormack scheme, was introduced into seismic wave modelling by Bayliss *et al.* (1986) and was implemented with an operator splitting. This 2–4 MacCormack splitting scheme has been used in a variety of seismic wave problems (Xie & Yao 1988; Tsingas *et al.* 1990; Vafidis *et al.* 1992; Dai *et al.* 1995).

Hixon (1997) developed several high-accuracy MacCormack schemes with the DRP/opt MacCormack scheme having the best trade-off between accuracy and computing efficiency. This scheme uses the DRP (Dispersion Relation Preserving) methodology of Tam and Webb (1993) for the spatial discretization and the 4–6 LDDRK (low dispersion and dissipation Runge–Kutta) scheme of Hu *et al.* (1996) for the time integration. Hixon (1998) demonstrated the superior performance of the DRP/opt scheme compared with the 2–4 scheme of Gottlieb & Turkel (1976). So we choose the DRP/opt MacCormack scheme to solve the elastic velocity-stress equations (eqs 7a–e) in curvilinear coordinates.

In MacCormack-type schemes, the central difference operator is split into forward and backward one-sided differences. The one-sided differences are alternately used in the stages of a multistage Runge–Kutta-type time integration method, and the central difference is recovered when the forward and backward differences are added together at the last time stage.

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4.1 One-sided difference operators

Hixon (1997) used the DRP (Dispersion Relation Preserving) scheme of Tam and Webb (1993) to optimize the dispersion error of the added central difference, and utilized the same methodology to optimize the dissipation error of the one-sided differences.

The forward and backward biased difference operators are

$$\widehat{W}_i^F = \frac{1}{\Delta x}\sum_{j=-1}^{3} a_j W_{i+j}, \tag{9a}$$

$$\widehat{W}_i^B = \frac{1}{\Delta x}\sum_{j=-1}^{3} -a_j W_{i-j}, \tag{9b}$$

where $\widehat{W}_i^F$ and $\widehat{W}_i^B$ denote the forward and backward difference operators with respect to ξ or η, $\{a_j\}$ are expansion coefficients and with values of

$$\begin{cases} a_{-1} = -0.30874, \\ a_0 = -0.6326, \\ a_1 = 1.2330, \\ a_2 = -0.3334, \\ a_3 = 0.04168. \end{cases} \tag{10}$$

These coefficients are obtained by minimizing the dissipation error at eight points or more per wavelength (Hixon 1997), and have formal fourth-order accuracy at the same time.

The above formulas of one-sided operators are defined for 1-D problem. For the current 2-D problem governed by eq. (8), we shall use the following hybrid one-sided difference operators:

$$\begin{aligned} \hat{L}^{FF} &= A\widehat{W}_{,\xi}^F + B\widehat{W}_{,\eta}^F, \\ \hat{L}^{BB} &= A\widehat{W}_{,\xi}^B + B\widehat{W}_{,\eta}^B, \\ \hat{L}^{FB} &= A\widehat{W}_{,\xi}^F + B\widehat{W}_{,\eta}^B, \\ \hat{L}^{BF} &= A\widehat{W}_{,\xi}^B + B\widehat{W}_{,\eta}^F, \end{aligned} \tag{11}$$

where $\widehat{W}_{,\xi}^F$ and $\widehat{W}_{,\xi}^B$ are the 1-D forward and backward difference operators with respect to ξ, $\widehat{W}_{,\eta}^F$ and $\widehat{W}_{,\eta}^B$ are the 1-D forward and backward difference operators with respect to η.

4.2 Time integration method

Having the split one-sided difference operators, the first-order partial differential velocity-stress equation system (eq. (8)) can be integrated by two-stage Runge–Kutta scheme or multistage Runge–Kutta schemes. Hixon (1997) and Hixon (1998) demonstrated the superior performance of the 4–6 LDDRK (Low-Dispersion and Low-Dissipation Runge–Kutta) scheme of Hu *et al.* (1996) compared with other Runge–Kutta schemes. The 4–6 LDDRK scheme uses two multistage Runge–Kutta schemes having four and six stages. The four- and six-stage RK schemes are alternately used to move to next time step.

The implementation of the DRP/opt MacCormack scheme and the 4–6 LDDRK scheme on the elastic equation (eq. 8) is as below:

Initial setting:

(i) generate 1-D operators $\widehat{W}_{,\xi}^F$, $\widehat{W}_{,\xi}^B$, $\widehat{W}_{,\eta}^F$ and $\widehat{W}_{,\eta}^B$, using eqs (9a and b)

(ii) form 2-D operators $\hat{L}^{FF}$, $\hat{L}^{BB}$, $\hat{L}^{FB}$ and $\hat{L}^{BF}$ by using eq. (11)

Time integration loop

(i) *sub time step 1*: four-stage RK time integration

$$\begin{aligned} h^{(1)} &= \Delta t\hat{L}^{FF}(W^n) \\ h^{(2)} &= \Delta t\hat{L}^{BB}\left(W^n + \alpha_2 h^{(1)}\right) \\ h^{(3)} &= \Delta t\hat{L}^{FF}\left(W^n + \alpha_3 h^{(2)}\right) \\ h^{(4)} &= \Delta t\hat{L}^{BB}\left(W^n + \alpha_4 h^{(3)}\right) \\ W^{n+1} &= W^n + \beta_1 h^{(1)} + \beta_2 h^{(2)} + \beta_3 h^{(3)} + \beta_4 h^{(4)}, \end{aligned} \tag{12}$$

(ii) *sub time step 2*: six-stage RK time integration

$$\begin{aligned} h^{(1)} &= \Delta t\hat{L}^{FB}(W^{n+1}) \\ h^{(2)} &= \Delta t\hat{L}^{BF}\left(W^{n+1} + \alpha_2 h^{(1)}\right) \\ h^{(3)} &= \Delta t\hat{L}^{FB}\left(W^{n+1} + \alpha_3 h^{(2)}\right) \\ h^{(4)} &= \Delta t\hat{L}^{BF}\left(W^{n+1} + \alpha_4 h^{(3)}\right) \\ h^{(5)} &= \Delta t\hat{L}^{FB}\left(W^{n+1} + \alpha_5 h^{(4)}\right) \\ h^{(6)} &= \Delta t\hat{L}^{BF}\left(W^{n+1} + \alpha_6 h^{(5)}\right) \\ W^{n+2} &= W^{n+1} + \beta_1 h^{(1)} + \beta_2 h^{(2)} + \beta_3 h^{(3)} \\ &\quad + \beta_4 h^{(4)} + \beta_5 h^{(5)} + \beta_6 h^{(6)}, \end{aligned} \tag{13}$$

(iii) *sub time step 3*: four-stage RK time integration

$$\begin{aligned} h^{(1)} &= \Delta t\hat{L}^{BB}(W^{n+2}) \\ h^{(2)} &= \Delta t\hat{L}^{FF}\left(W^{n+2} + \alpha_2 h^{(1)}\right) \\ h^{(3)} &= \Delta t\hat{L}^{BB}\left(W^{n+2} + \alpha_3 h^{(2)}\right) \\ h^{(4)} &= \Delta t\hat{L}^{FF}\left(W^{n+2} + \alpha_4 h^{(3)}\right) \\ W^{n+3} &= W^{n+2} + \beta_1 h^{(1)} + \beta_2 h^{(2)} + \beta_3 h^{(3)} + \beta_4 h^{(4)}, \end{aligned} \tag{14}$$

(iv) *sub time step 4*: six-stage RK time integration

$$\begin{aligned} h^{(1)} &= \Delta t\hat{L}^{BF}(W^{n+3}) \\ h^{(2)} &= \Delta t\hat{L}^{FB}\left(W^{n+3} + \alpha_2 h^{(1)}\right) \\ h^{(3)} &= \Delta t\hat{L}^{BF}\left(W^{n+3} + \alpha_3 h^{(2)}\right) \\ h^{(4)} &= \Delta t\hat{L}^{FB}\left(W^{n+3} + \alpha_4 h^{(3)}\right) \\ h^{(5)} &= \Delta t\hat{L}^{BF}\left(W^{n+3} + \alpha_5 h^{(4)}\right) \\ h^{(6)} &= \Delta t\hat{L}^{FB}\left(W^{n+3} + \alpha_6 h^{(5)}\right) \\ W^{n+4} &= W^{n+3} + \beta_1 h^{(1)} + \beta_2 h^{(2)} + \beta_3 h^{(3)} \\ &\quad + \beta_4 h^{(4)} + \beta_5 h^{(5)} + \beta_6 h^{(6)}, \end{aligned} \tag{15}$$

criteria for stopping

end of time integration.

In the above equations, the coefficients of the four-stage scheme are:

$$\begin{array}{ll} \alpha_1 = 0 & \beta_1 = 1/6 \\ \alpha_2 = 0.5 & \beta_2 = 1/3 \\ \alpha_3 = 0.5 & \beta_3 = 1/3 \\ \alpha_4 = 1 & \beta_4 = 1/6 \end{array}. \tag{16}$$

The coefficients of the six-stage scheme are:

$$\begin{array}{ll} \alpha_1 = 0 & \beta_1 = 0.0467621 \\ \alpha_2 = 0.353323 & \beta_2 = 0.137286 \\ \alpha_3 = 0.999597 & \beta_3 = 0.170975 \\ \alpha_4 = 0.152188 & \beta_4 = 0.197572 \\ \alpha_5 = 0.534216 & \beta_5 = 0.282263 \\ \alpha_6 = 0.603907 & \beta_6 = 0.165142 \end{array}. \tag{17}$$

These coefficients are obtained by minimizing the dispersion and dissipation error of the scheme while keeping formally fourth-order

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accuracy (Hu *et al.* 1996). It should be noted that the order of the one-sided difference operators is interchanged every step to avoid numerical biasing in each of the multistage RK schemes, and the four-stage and six-stage RK schemes are alternately used to suppress both dissipation and dispersion errors of the traditional Runge–Kutta scheme.

5 IMPLEMENTATION OF FREE SURFACE CONDITION WITH SURFACE TOPOGRAPHY

At the free surface, the difference operators should take the free surface boundary condition into account. Fig. 2 shows the computational grid points near the free surface which can be classified into three categories:

(i) interior grid points: located three grid points away from the free surface (i.e. points A and below).

(ii) boundary grid points: located between the interior grid points and the free surface (i.e. points B, C and D).

(iii) ghost grid points: located above the free surface.

The DRP/opt MacCormack scheme can be directly used to update the velocity-stress equations (eqs (7a–e)) on the interior grid points. On the boundary grid points, calculating the derivatives on the right-hand side of eqs (7a–e) requires values of stress and velocity components on the ghost grid points. These values should be specified in such a way that the free surface boundaries condition can be preserved.

In this section, we describe the method to preserve the free boundary condition in calculating the derivatives of the stress and velocity components on the boundary grid points.

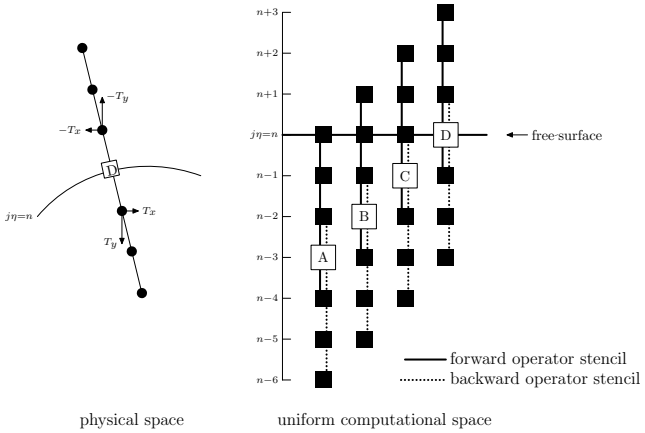


Figure 2. The grid stencil of seven points DRP/opt MacCormack scheme for Traction Image on boundary conforming grids. The left panel shows the boundary-conforming grid in the physical space, and the right panel shows the equivalent uniform grid in the computational space. The free surface is set to be at $j_\eta = n$ layer. Grid points from 1 to $n-3$ layers are called interior grid points, as points A and below; grid points from $n-2$ to n layers are called boundary grid points, as points B, C and D; grid points above the free surface, from $n+1$ to $n+3$ layers, are called ghost grid points. To update the velocity-stress equations on boundary grid points (points B, C and D), values on the ghost points are required. To satisfy the traction-free boundary conditions, the values of normal traction components T_x and T_y on the ghost points are set antisymmetrically about the free surface.

5.1 Physical free surface boundary condition at irregular surfaces

In the presence of surface topography, the free surface boundary condition requires the traction T over the free surface to be zero:

$$T = \vec{n} \cdot \sigma = 0. \tag{18}$$

Substituting the specific expression of normal vector, eq. (5), into above equation yields

$$T_x = (\eta_{,x}\sigma_{xx} + \eta_{,y}\sigma_{xy})\frac{1}{|\nabla\eta(x,y)|} = 0, \tag{19a}$$

$$T_y = (\eta_{,x}\sigma_{xy} + \eta_{,y}\sigma_{yy})\frac{1}{|\nabla\eta(x,y)|} = 0. \tag{19b}$$

After differentiating the above equations with respect to time and then inserting eqs (7c–e) in them, we can obtain the following equation

$$X\begin{bmatrix} v_{x,\eta} \\ v_{y,\eta} \end{bmatrix} = -Y\begin{bmatrix} v_{x,\xi} \\ v_{y,\xi} \end{bmatrix}, \tag{20}$$

where

$$X = \begin{bmatrix} \chi\eta_{,x}\eta_{,x} + \mu\eta_{,y}\eta_{,y} & \lambda\eta_{,x}\eta_{,y} + \mu\eta_{,y}\eta_{,x} \\ \lambda\eta_{,y}\eta_{,x} + \mu\eta_{,x}\eta_{,y} & \chi\eta_{,y}\eta_{,y} + \mu\eta_{,x}\eta_{,x} \end{bmatrix},$$

$$Y = \begin{bmatrix} \chi\eta_{,x}\xi_{,x} + \mu\eta_{,y}\xi_{,y} & \lambda\eta_{,x}\xi_{,y} + \mu\eta_{,y}\xi_{,x} \\ \lambda\eta_{,y}\xi_{,x} + \mu\eta_{,x}\xi_{,y} & \chi\eta_{,y}\xi_{,y} + \mu\eta_{,x}\xi_{,x} \end{bmatrix}.$$

Eq. (20) can be further solved as follows

$$\begin{bmatrix} v_{x,\eta} \\ v_{y,\eta} \end{bmatrix} = -X^{-1}Y\begin{bmatrix} v_{x,\xi} \\ v_{y,\xi} \end{bmatrix}. \tag{21}$$

This equation links the derivatives of the velocity along the longitude (ξ) and latitude (η) directions, respectively, in general curvilinear coordinates, which provides a constraint on the velocity components at the free surface.

5.2 Traction-imaging for traction derivatives

By boundary-conforming transformation, the physically irregular topography is transformed into a flat surface in the computational space. At this 'flat' surface, the traction-free boundary conditions, eqs (19a and b), should be preserved.

In the transformed computational space, the free surface boundary is set to be at $j_\eta = n$ (see Fig. 2). Similar to the Stress Image method for planar surface in physical space, if we set the values of traction on the ghost grid points to be antisymmetric about the free surface as follows,

$$T_x|_{j_\eta=n+j} = -T_x|_{j_\eta=n-j}, \tag{22a}$$

$$T_y|_{j_\eta=n+j} = -T_y|_{j_\eta=n-j}, \tag{22b}$$

$$T_x|_{j_\eta=n} = T_y|_{j_\eta=n} = 0. \tag{22c}$$

We can ensure the traction-free boundary conditions, eqs (19a and b), to be satisfied at the free surface. Because the quantities being anti-symmetrically reflected onto the ghost grid points are traction, we named this method as Traction Image method.

This traction-imaging technique can not be directly used in eqs (7a and b), the momentum equations in curvilinear coordinates,

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because the variables differentiated with respect to η are stress components instead of traction components. However, by using the conservative form of ∇ operator (Thompson *et al.* 1985), we can rewrite the momentum equations into a conservative form (see Appendix B), an alternative form of the eqs (7a and b) (here and throughout this paper, the word 'conservative' means the conservative form of ∇ operator),

$$\rho v_{x,t} = \frac{1}{J}\{[J(\xi_{,x}\sigma_{xx} + \xi_{,y}\sigma_{xy})]_{,\xi} + [J(\eta_{,x}\sigma_{xx} + \eta_{,y}\sigma_{xy})]_{,\eta}\}, \tag{23a}$$

$$\rho v_{y,t} = \frac{1}{J}\{[J(\xi_{,x}\sigma_{xy} + \xi_{,y}\sigma_{yy})]_{,\xi} + [J(\eta_{,x}\sigma_{xy} + \eta_{,y}\sigma_{yy})]_{,\eta}\}. \tag{23b}$$

In the above equations, the terms differentiated with respect to η happen to be the components of normal traction multiplied by a time-independent quality $J|\nabla\eta|$. The Jacobian J and the quality $|\nabla\eta|$ are non-singular and positive, and can be extended to the upper computational space above the free surface via a symmetric continuation, thus the settings of traction-imaging on ghost grid points (eqs 22a and b) can be modified to the following formulas,

$$(\eta_{,x}\sigma_{xx} + \eta_{,y}\sigma_{xy})|_{j_\eta=n+j} = -(\eta_{,x}\sigma_{xx} + \eta_{,y}\sigma_{xy})|_{j_\eta=n-j}, \tag{24a}$$

$$(\eta_{,x}\sigma_{xy} + \eta_{,y}\sigma_{yy})|_{j_\eta=n+j} = -(\eta_{,x}\sigma_{xy} + \eta_{,y}\sigma_{yy})|_{j_\eta=n-j}. \tag{24b}$$

And eq. (22c) is guaranteed by eq. (21), that is, the velocity constraint at the free surface.

With the above modified traction-imaging formulas, we can easily update the momentum equation in conservative form (eqs 23a and b) on the boundary grid points with DRP/opt MacCormack scheme while preserving the traction-free boundary conditions.

For a planar surface in physical space and adapting the Cartesian coordinate with $\eta_{,x} = 0$, $\eta_{,y} = 1$ and $J = 1$, the traction-imaging settings, eqs (24a and b), are reduced to

$$\sigma_{yx}|_{j_y=n+j} = -\sigma_{yx}|_{j_y=n-j}, \tag{25a}$$

$$\sigma_{yy}|_{j_y=n+j} = -\sigma_{yy}|_{j_y=n-j}. \tag{25b}$$

This is exactly the stress-imaging settings in the conventional Stress Image method for planar surface (Levander 1988; Graves 1996). We can see that our Traction Image method is more general than the conventional Stress Image method.

5.3 Compact scheme for velocity derivatives

The Traction Image method proposed above provides all required values on the ghost grid points to calculate traction derivatives on the boundary grid points. However, the situation for the velocity derivatives in eqs (7c–e) is different. The free surface boundary conditions require the velocity components to satisfy the derivative relations of eq. (20) at the free surface, but this constraint is not enough to uniquely set all required values of velocity on the ghost grid points. So we should find a solution to calculate the derivatives of velocity on the boundary grid points without using values on ghost grid points.

Before going further, two points are worthy to note. First, all the derivatives with respect to ξ on the boundary grid points can be calculated by using interior finite difference scheme without any problem. Thus for the grid points located on the free surface, as D in Fig. 3, the velocity derivatives with respect to η can be obtained by using eq. (21). Second, point A in Fig. 3 is a interior grid point, so $v_{x,\eta}|_A$ and $v_{y,\eta}|_A$ can be calculated by using interior finite difference scheme.

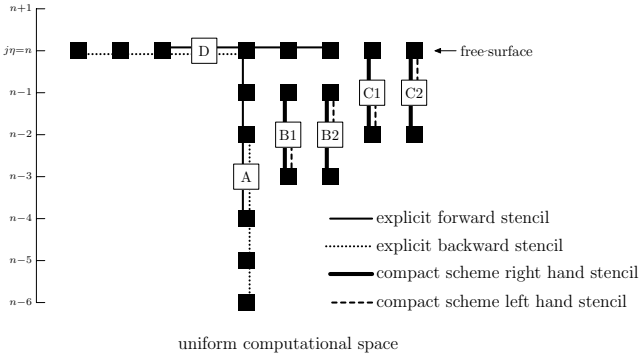


Figure 3. The grid stencil of DRP/opt MacCormack scheme and 4/4 compact MacCormack scheme to calculate normal derivatives of velocity on the boundary grid points. For grid points located on the free surface, as grid D, $\mathbf{v}_{,\eta}$ can be derived from $\mathbf{v}_{,\xi}$ according to eq. (21); For interior grid point A, $\mathbf{v}_{,\eta}|_A$ can be calculated by the interior difference scheme; For grid points B1 and C1, or B2 and C2, the 4/4 compact MacCormack scheme is used, which needs values of the velocity components and its derivatives only on three grid points.

Thus, provided with the derivatives of velocity with respect to η on grid points A and D, we can use a compact scheme to calculate the derivatives of velocity on the other boundary grid points, such as grid points B1 and C1 or grid points B2 and C2 in Fig. 3. We choose the following 4/4 compact MacCormack scheme (Hixon & Turkel (2000)) to calculate the derivatives of velocity in eqs (7c–e) on the boundary grid points,

$$\widehat{W}^B_{j-1,\eta} + 2\widehat{W}^B_{j,\eta} = \frac{1}{2\Delta x}\left\{W_{j+1} + 4W_j - 5W_{j-1}\right\}, \tag{26a}$$

$$2\widehat{W}^F_{j,\eta} + \widehat{W}^F_{j+1,\eta} = \frac{1}{2\Delta x}\left\{5W_{j+1} - 4W_j - W_{j-1}\right\}, \tag{26b}$$

where $\widehat{W}^F_{j,\eta}$ and $\widehat{W}^B_{j,\eta}$ denote, respectively, the forward and backward difference operators of the velocity components with respect to η at grid j.

6 OTHER NUMERICAL TREATMENTS

6.1 Interior interface conditions

Finite difference schemes are suitable for any kind of voluminal heterogeneity. Following Moczo *et al.* (2002), we treat the discontinuous interior interfaces by effective parameters:

The density is calculated by arithmetic average:

$$\rho_{ij} = \frac{1}{\Delta S}\int_{i-1/2}^{i+1/2}\int_{j-1/2}^{j+1/2}\rho\, dx\, dy. \tag{27}$$

The Lame parameters are calculated by harmonic average:

$$\frac{1}{\mu_{ij}} = \frac{1}{\Delta S}\int_{i-1/2}^{i+1/2}\int_{j-1/2}^{j+1/2}\frac{1}{\mu}\, dx\, dy, \tag{28}$$

$$\frac{1}{\lambda_{ij}} = \frac{1}{\Delta S}\int_{i-1/2}^{i+1/2}\int_{j-1/2}^{j+1/2}\frac{1}{\lambda}\, dx\, dy. \tag{29}$$

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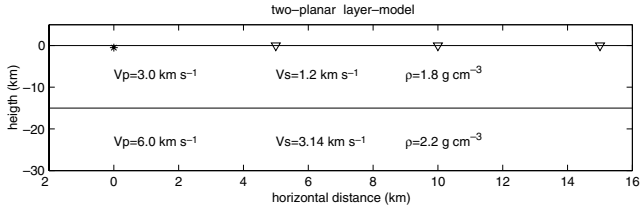


Figure 4. Model of a two-planar layer. The media interface is at 15 km; the source locates at (0, −0.5), which is marked as an asterisk; the three selected receivers for detailed comparisons locate at (5, 0), (10, 0), and (15, 0) and are indicated by triangle symbols.

With these effective parameters, the interface discontinuities can be converted to voluminal heterogeneities which are feasible for finite difference solvers.

6.2 Numerical source

The velocity-stress equations with source terms are

$$\rho v_{i,t} = [\nabla \cdot \sigma]_i + f_i, \tag{30a}$$

$$\sigma_{ij,t} = \left[E : \frac{1}{2}(\nabla \vec{v} + \vec{v}\nabla) \right]_{ij} + s_{ij}. \tag{30b}$$

Where, f_i and s_{ij} are the source terms and have forms:

$$f_i = -s(t) \sum_{j,k=1}^{2} M_{ji} \frac{\partial \xi_k}{\partial x_j} \frac{\partial}{\partial \xi_k} \delta(\mathbf{x} - \mathbf{x}_s), \tag{30c}$$

$$s_{ij} = -\dot{s}(t) M_{ij} \delta(\mathbf{x} - \mathbf{x}_s). \tag{30d}$$

Where, $\xi_k \in \{\xi, \eta\}$, $x_j \in \{x, y\}$; M_{ij} is the ijth component of source moment tensor; $s(t)$ and $\dot{s}(t)$ are the source time function and its derivative with respect to time; $\mathbf{x}_s$ is the coordinate of the source. The δ function is approximated by Herrmann pseudo-delta functions (Herrmann 1979; Wang *et al.* 2001).

6.3 Absorbing boundary

We use the split-field perfectly matched layer (PML) approach (Berenger 1994; Marcinkovich & Olsen 2003) to absorb the wavefield around the computational grids.

7 NUMERICAL VALIDATIONS

In order to validate the Traction Image method, we consider six numerical tests: a two-planar-layer model (Fig. 4) to validate that the DRP/opt MacCormack scheme can be used to solve the first-order partial differential velocity-stress equations on non-staggered grids; a homogeneous half-space model with topography of hill (Fig. 5), canyon (Fig. 6), and both canyon and hill (Fig. 7) to examine the Traction Image method on smooth boundaries; a semi-circular model (Fig. 8) to test the ability of the method for non-smooth topography; and a step shape model (Fig. 9) to further test the range of applicability of the method for extreme topography. Among the

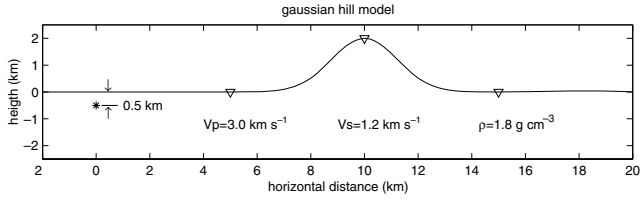


Figure 5. Model of a half-space with Gaussian shape hill topography. The shape of the hill is defined by a Gaussian function $y = h \exp^{-(x-x_0)^2/a^2}$, with $h = 2$, $a = 3$, and $x_0 = 10$; the source locates at (0, −0.5), which is marked as an asterisk; the three selected receivers for detailed comparisons locate at (5, 0), (10, 2), and (15, 0).

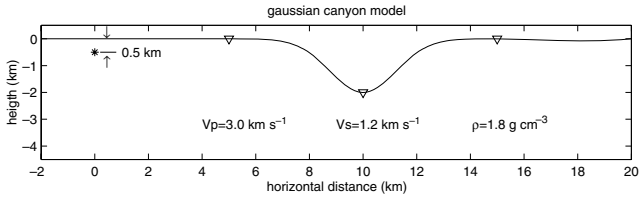


Figure 6. Model of a half-space with Gaussian shape canyon topography. The shape of the canyon is defined by a Gaussian function $y = -h \exp^{-(x-x_0)^2/a^2}$, with $h = 2$, $a = 3$, and $x_0 = 10$; the source locates at (0, −0.5), which is marked as an asterisk; the three selected receivers for detailed comparisons locate at (5, 0), (10, −2), and (15, 0).

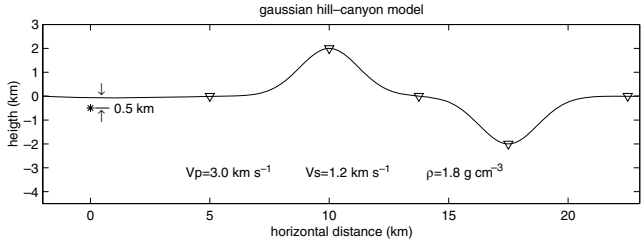


Figure 7. Model of a half-space with hill-canyon shape topography. The shape of the surface is defined by a Gaussian function $y = h \exp^{-(x-x_0)^2/a^2} - h \exp^{-(x-x_1)^2/a^2}$, with $h = 2$, $a = 3$, $x_0 = 10$, and $x_1 = 17.5$; the source locates at (0, −0.5), which is marked as an asterisk; the five selected receivers for detailed comparisons locate at (5, 0), (10, 2), (13.75, 0), (17.5, −2), and (22.5, 0).

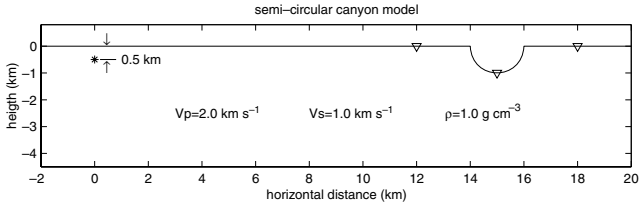


Figure 8. Model of a half-space with semi-circular canyon shape topography. The circular centre locates at (15, 0), and the radius is 1 km; the source locates at (0, −0.5), which is marked as an asterisk; the three selected receivers for detailed comparisons locate at (12, 0), (15, −1), (18, 0).

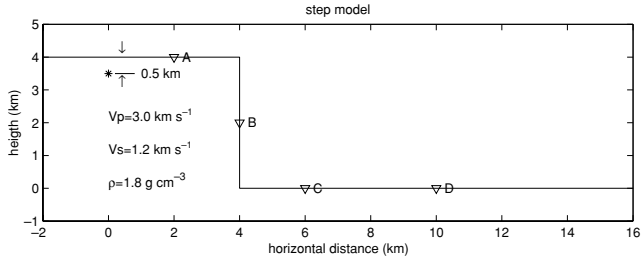


Figure 9. Model of a half-space with step shape topography. The step descends from 4 km to 0 km at $x = 4$, the source locates at (0, 3.5), which is marked as an asterisk; the four selected receivers for detailed comparisons locate at A (2, 4), B (4, 2), C (6, 0), and D (10, 0).

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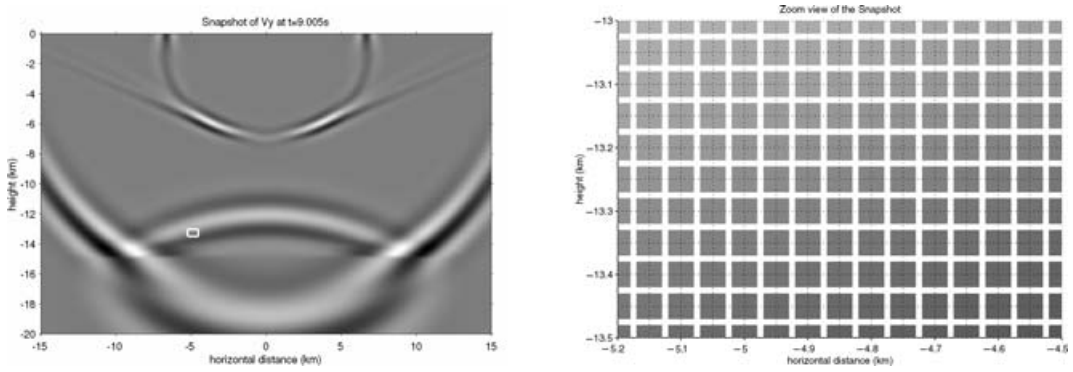


Figure 10. The snapshot of v_y at about $t = 9$ s of the two-planar-layer model. Right panel is the zoom view of the white square area on the left panel. On the right panel, the intersections between the dotted lines indicate the locations of the grid points.

tests, we compare our simulated results of two-planar-layer model with those from the generalized reflection/transmission coefficients method (Kennett 1983; Luco & Apsel 1983; Chen 1999), and compare our results of other tests with those from the boundary element method (Brebbia 1981; Dominguez 1993; Ge *et al.* 2005).

In all tests, an explosive point source buried 0.5 km below the free surface is used and the location is indicated by the asterisks in Figs 4–9. For the semi-circular model, the source time function is a Ricker wavelet with centre frequency of 1.0 Hz (thus the maximum frequency is about 2.5 Hz) and time-shift of 3.0 s; for others models, the source time function is a Ricker wavelet with centre frequency of 0.96 Hz (thus the maximum frequency is about 2.4 Hz) and time-shift of 3.125 s.

7.1 Results for a two-planar-layer model

The first test model considered here consists of two planar layers with the parameters as shown in Fig. 4. In this test, the grid spacing is 0.05 km and the number of grid points per shortest shear wavelength is about 10.

The wavefield snapshot of vertical velocity v_y at about 9 s was shown in Fig. 10. Hustedt *et al.* (2004) showed that second-order

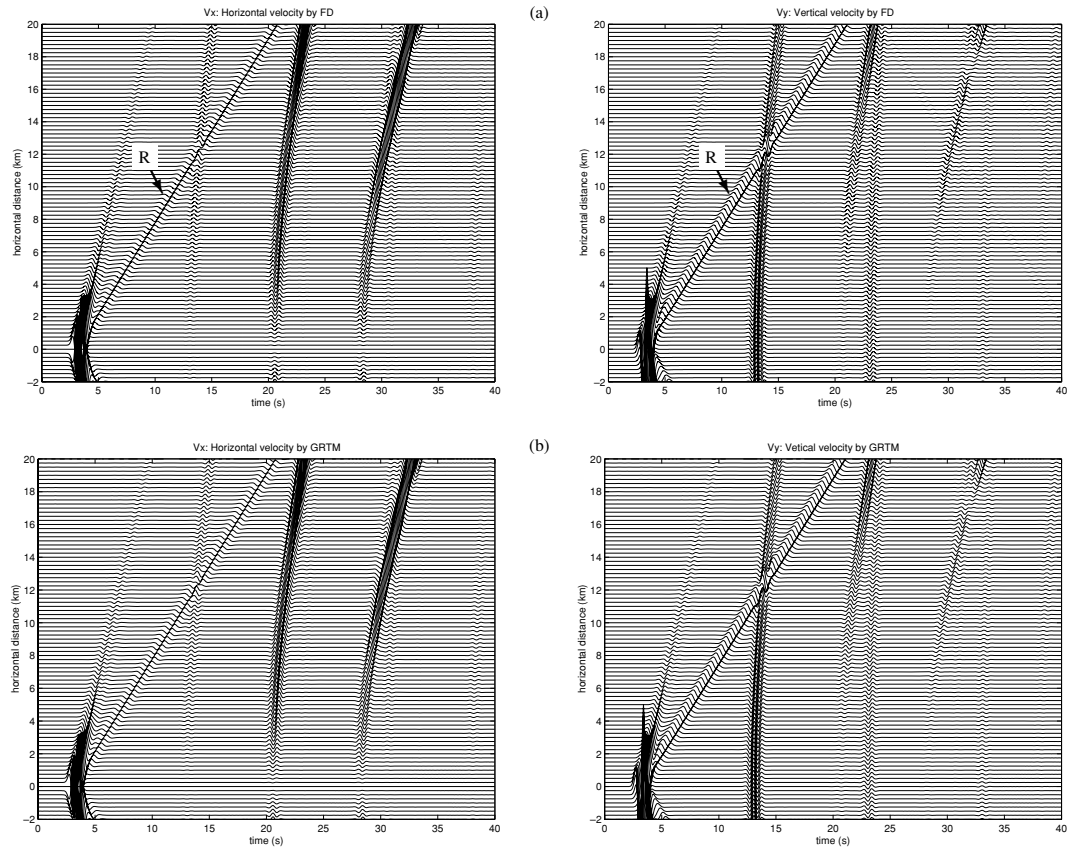


Figure 11. Trace records of particle velocity along the free surface of the two-planar-layer model. Left panel shows the horizontal velocity component, v_x; right panel shows the vertical velocity component, v_y. (a) Obtained by finite difference method, (b) Obtained by the generalized reflection/transmission coefficient method. In (a), arrow *R* indicates Rayleigh phase.

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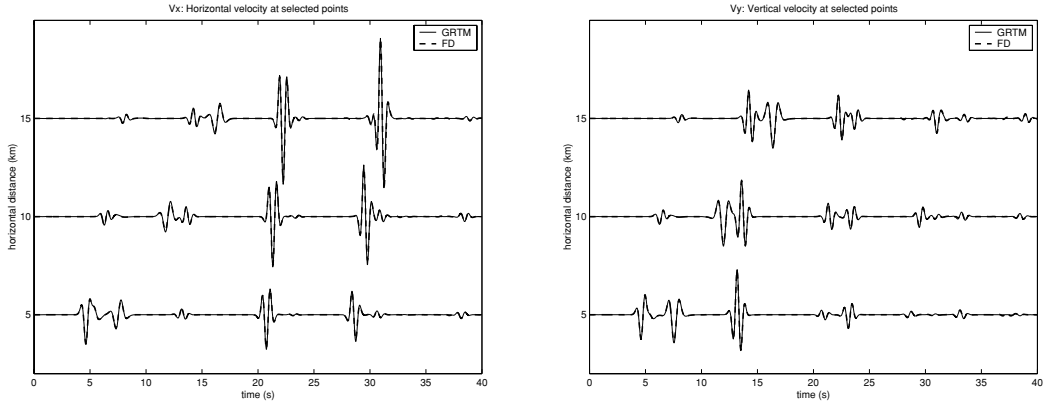


Figure 12. Detailed comparisons of the seismograms calculated with the finite difference (dashed line) and those from the generalized reflection/transmission coefficient method (solid line) of the two-planar-layer model at the selected observation points as indicated in Fig. 4. The amplitudes are five times enlarged compared with Fig. 11.

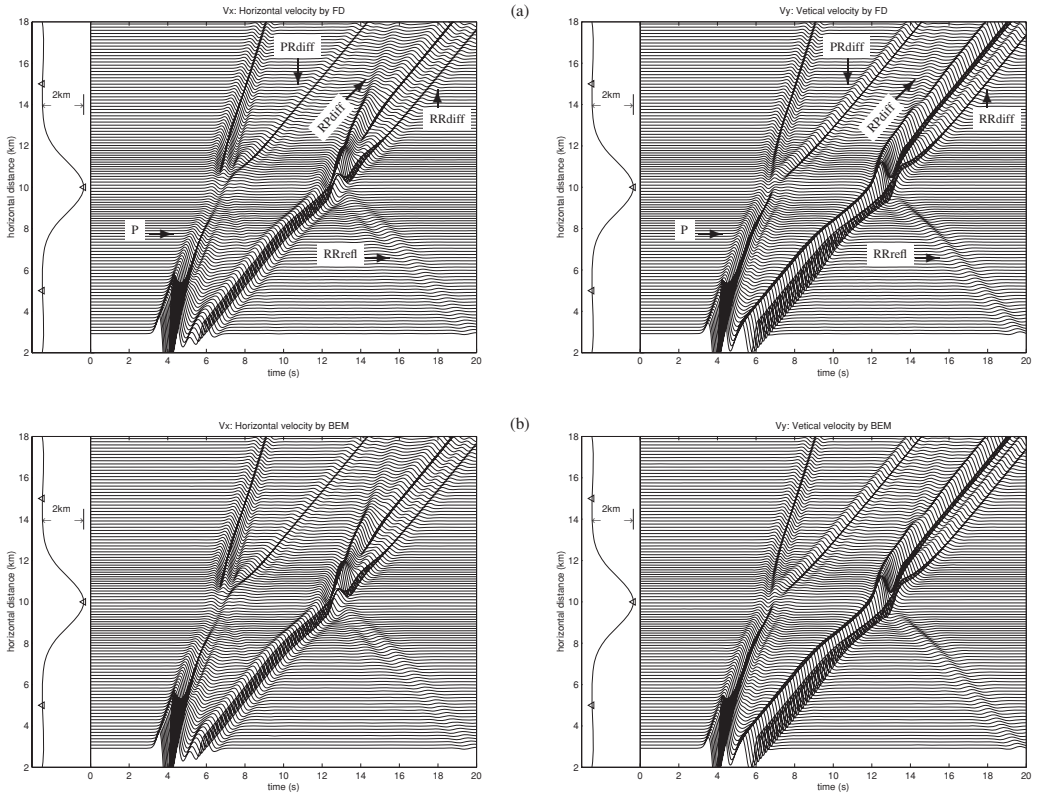


Figure 13. Trace records of the velocity components along the free surface of the Gaussian hill shape model. Left panel shows the horizontal velocity component, v_x; right panel shows the vertical velocity component, v_y. (a) Obtained by finite difference method, (b) Obtained by the boundary element method. Arrow *P* indicates the incident *P* wave; others arrows mean the conversion: *P* diffracted to Rayleigh (PRdiff), Rayleigh diffracted to *P* (RPdiff), Rayleigh diffracted to Rayleigh (RRdiff), Rayleigh reflected to Rayleigh (RRrefl).

non-staggered central finite difference for the first derivative causes two uncoupled staggered grids: one that is not excited while the other is related to the source position when source is excited at one single grid node. Komatitsch *et al.* (1996) found that non-staggered central finite difference yielded numerical artefacts in the case of high velocity contrasts at an interface. These phenomena do not exist in Fig. 10, which demonstrates that the non-staggered DRP/opt MacCormack scheme can accurately solve the first-order velocity-stress equations.

To check the accuracy of the scheme, we compared the results with those obtained by the generalized reflection/transmission coefficient method (Chen 1993, 1999). Fig. 11 shows the comparisons of the synthetic seismograms on the surface. Fig. 12 displays the details of the comparisons at the selected observation points indicated in Fig. 4 by the triangle symbols. The seismograms are calculated up to 40 s with time step of $\Delta t = 0.005$ s.

These comparisons show an excellent agreement with all phases including Rayleigh phases. The Rayleigh phase is mainly caused by

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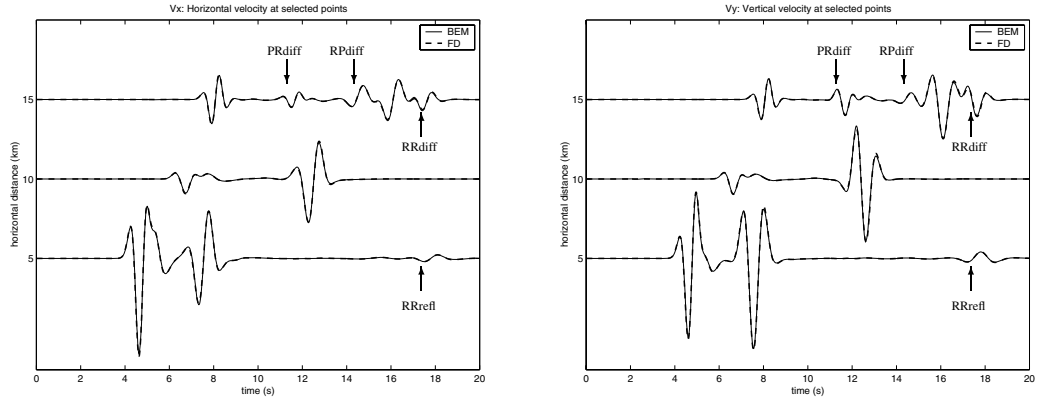


Figure 14. Detailed comparisons between the seismograms calculated with the finite difference (dashed line) and those from the boundary element method (solid line) of the Gaussian hill shape model at the selected observation points as indicated in Fig. 5. The amplitudes are five times enlarged compared with Fig. (13).

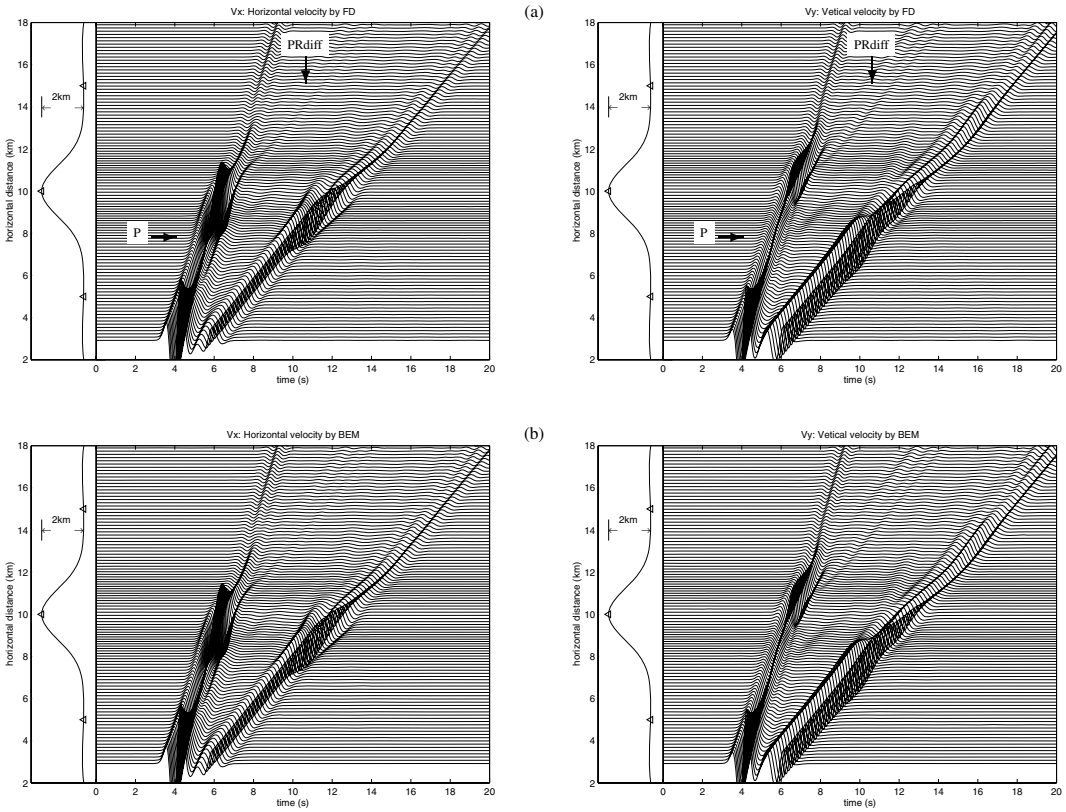


Figure 15. Trace records of the velocity components along the free surface of the Gaussian canyon shape model. Left-hand side panel shows the horizontal velocity component, v_x; Right-hand side panel shows the vertical velocity component, v_y. (a) Obtained by finite difference method, (b) Obtained by the boundary element method. Arrow *P* indicates the incident *P* wave, and RPdiff indicates the Rayleigh phase diffracted from P.

the presence of free surface, and it can be used to justify the accuracy of the numerical free boundary condition treatment. According to Bayliss *et al.* (1986), the Rayleigh wave cannot be simulated as accurate as the body waves when the free surface boundary condition is treated by the Characteristic Variables method, the most popular method in non-staggered finite difference simulation. However, the excellent agreement between our results with those by the accurate generalized reflection/transmission coefficient method for both body waves and Rayleigh waves demonstrates that the Traction Image method proposed in this study can accurately treat the free surface boundary condition. In the following sections, we shall further demonstrate the validity of our new method in the case with irregular shape free surface.

7.2 Results for a half-space with Gaussian shape hill topography

The first model with irregular topography considered here is a half-space with Gaussian shape hill. The shape of the hill is defined by a Gaussian function $y = h \exp^{-(x-x_0)^2/a^2}$, with $h = 2$ km, $a = 3$ km, and $x_0 = 10$ km; the medium parameters are shown in the model sketch (Fig. 5). The grid spacing along the surface is about

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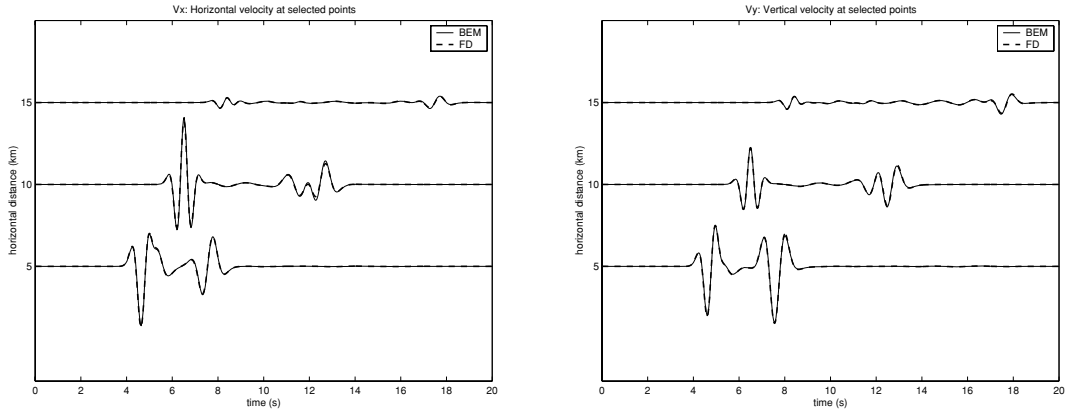


Figure 16. Detailed comparisons between the seismograms calculated with the finite difference (dashed line) and those from the boundary element method (solid line) of the Gaussian canyon shape model at the selected observation points as indicated in Fig. 6. The amplitudes are four times enlarged compared with Fig. (13).

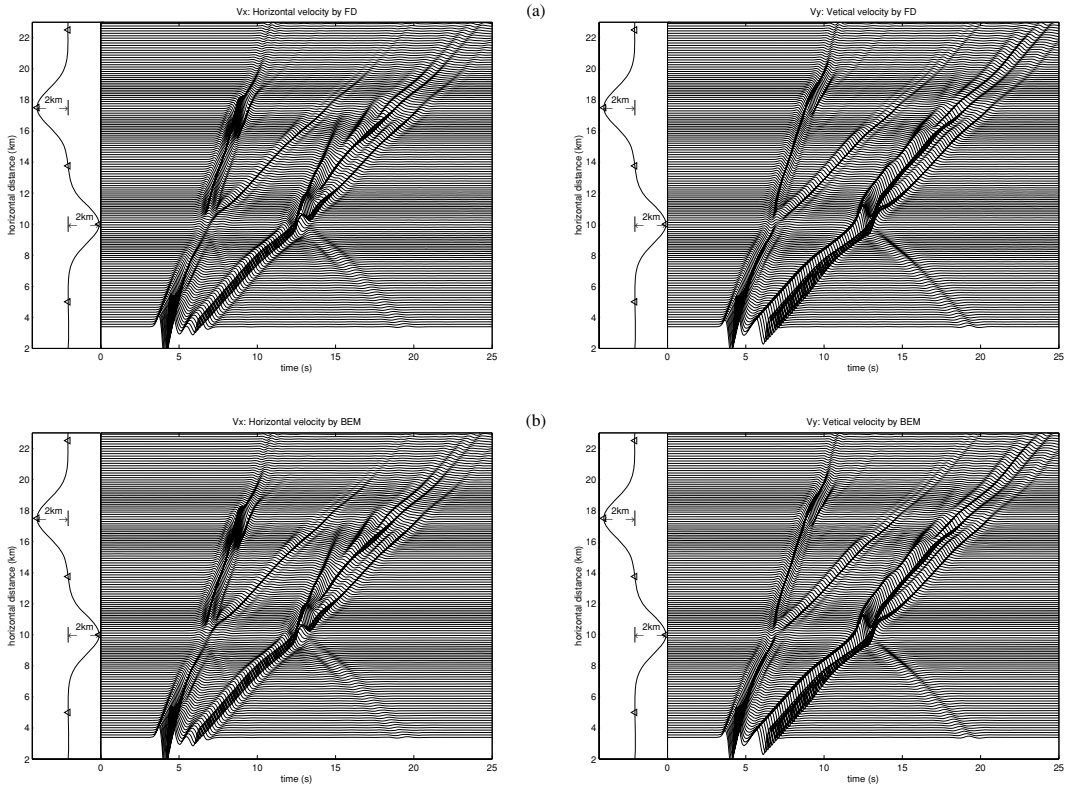


Figure 17. Trace records of the velocity components along the free surface of the Gaussian hill-canyon shape model. Left panel shows the horizontal velocity component, v_x; right panel shows the vertical velocity component, v_y. (a) Obtained by finite difference method, (b) Obtained by the boundary element method.

0.051 km and the number of grid points per shortest shear wavelength is about 10. All seismograms shown here are calculated up to 20 s with time step of $\Delta t = 0.005$ s.

In this test, the results obtained by BEM (Ge *et al.* 2005) are taken as reference. Comparisons of the synthetic seismograms for receivers on the surface are shown in Fig. 13. The agreement between the results calculated by the two methods are excellent. The Rayleigh wave diffracted by the apex of the hill from the *P* wave, and the reflected and converted Rayleigh waves from the incident Rayleigh wave, can be clearly seen in both the horizontal and the vertical components. Moreover, a Rayleigh-to-*P* conversion phase can be recognized, especially in the horizontal component.

Fig. 14 shows the comparisons of the detailed features of the synthetic seismograms by the two methods for three representative receivers indicated by triangle symbols in Fig. 5. Although the amplitudes of wavefields shown have been amplified five times compared with those in Fig. 13, the excellent agreement still be seen.

7.3 Results for a half-space with Gaussian shape canyon topography

The second model with irregular topography considered is a half-space with Gaussian shape canyon topography. The shape of the canyon is defined by a Gaussian function $y = -h \exp^{-(x-x_0)^2/a^2}$,

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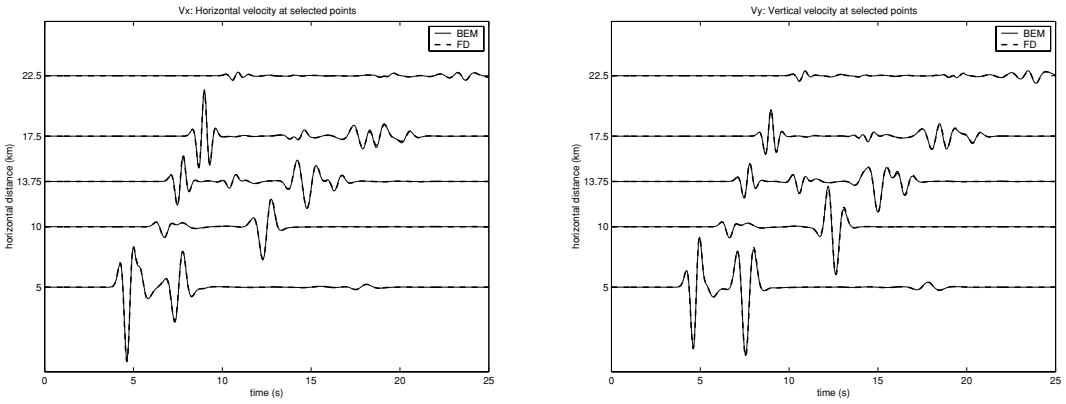


Figure 18. Detailed comparisons between the seismograms calculated with the finite difference (dashed line) and those from the boundary element method (solid line) of the Gaussian hill-canyon shape model at the selected observation points as indicated in Fig. 7. The amplitudes are five times enlarged compared with Fig. (17).

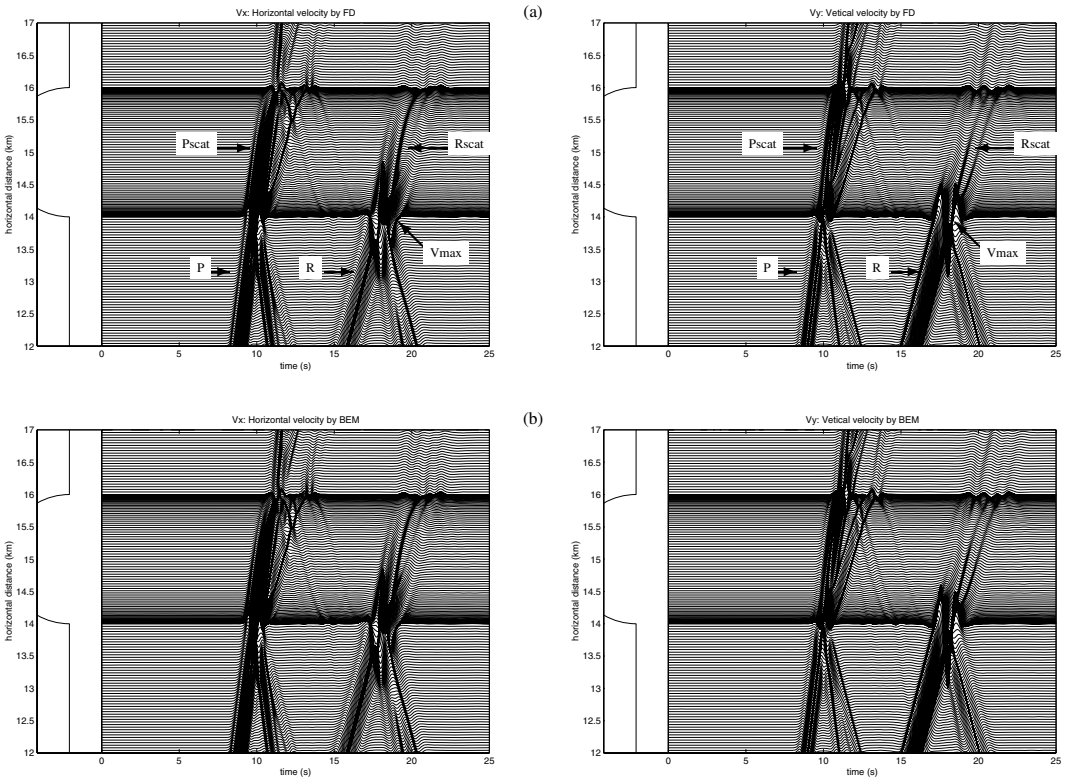


Figure 19. Trace records of the velocity components along the free surface of the semi-circular canyon shape model. Left-hand side panel shows the horizontal velocity component, v_x; right-hand side panel shows the vertical velocity component, v_y. (a) Obtained by finite difference method, (b) Obtained by the boundary element method. Arrow *P* and *R* indicate the incident *P* and Rayleigh wave, Pscat and Rscat indicate the forward scattering *P* and Rayleigh wave. Arrow Vmax indicates the largest amplitude of the ground motion.

with $h = 2$, $a = 3$, and $x_0 = 10$ km; the medium parameters are shown in the model sketch (Fig. 6). The grid spacing is same as those for the previous hill model.

Fig. 15 shows the comparisons of the synthetic seismograms calculated by our finite difference method and by BEM for receivers located on the surface. Fig. 16 shows some of the detailed features of the waveform comparisons at three representative receivers as indicated by triangle symbols in Fig. 6. The seismograms shown in both figures are calculated up to 20 s with time step of $\Delta t = 0.005$ s. Again, we can see an excellent agreement between the results from both methods for this canyon topography model, which confirms the validity of our method at least for the case with smooth canyon topography.

It can be seen that the smooth canyon significantly blocks the propagation of body wave and less significantly blocks the Rayleigh wave compared with the hill topography. Another notable feature is the scattered waves by the canyon topography are too weak to be observed.

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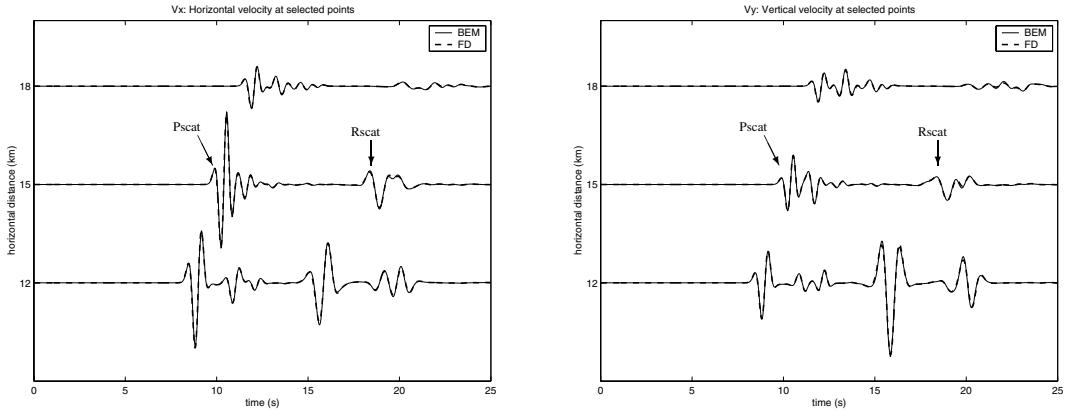


Figure 20. Detailed comparisons between the seismograms calculated with the finite difference (dashed line) and those from the boundary element method (solid line) of the semi-circular canyon shape model at the selected observation points as indicated in Fig. 8. The amplitudes are five times enlarged compared with Fig. (19).

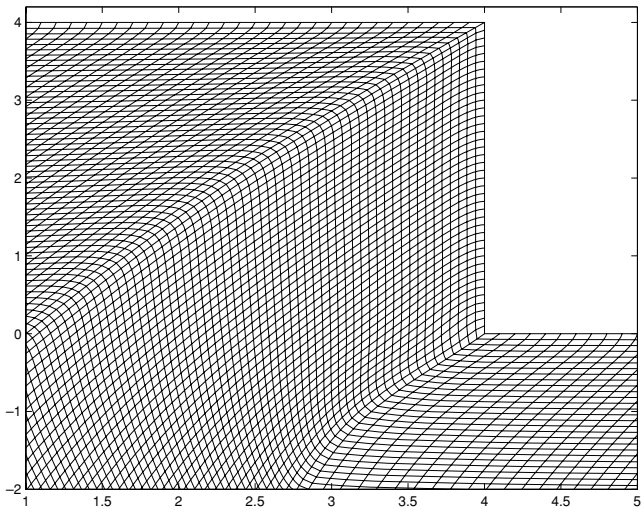


Figure 21. The grids near the step corner in the step shape model. For clarity, the grids are displayed with a reducing density factor of 2.

7.4 Results for a half-space with hill-canyon shape topography

The third model with irregular topography consists of a half-space with hill-canyon shape topography as illustrated in Fig. 7, in which the specific model parameters are given. The grid spacing is about 0.052 km and the number of grid points per shortest shear wavelength is about 10. The seismograms are calculated up to 20 s with time step of $\Delta t = 0.005$ s.

Comparisons of synthetic seismograms for this model are shown in Fig. 17 for the receivers on the surface. The detailed features of the comparisons at five representative observation points are shown in Fig. 18, in which the amplitudes of seismograms have been enlarged by five times compared with Fig. 17. An excellent agreement between the results from two methods can be clearly seen, though for such a complex topography model.

From the above computed results, we can see strong and complex scattered waves on the surface between the hill and canyon, which may imply a severe seismic hazard in such complex topographic area.

7.5 Results for a half-space with semi-circular shape canyon topography

In this section, we shall test our method with a semi-circular canyon model as illustrated in Fig. 8. All the previous topography models we have considered are smooth ones, that is, with continuous and finite slopes everywhere. The semi-circular canyon considered here, however, is an extreme topography model in which the vertical/horizontal shape ratio of the canyon is quite large (1/2) and the slopes of canyon's edges ($x = 14$ and 16 km) tend to infinity. Testing such an extreme model also helps to further validate the accuracy and the range of applicability of our method.

Comparisons of the synthetic seismograms for receivers distributed along the surface are shown in Fig. 19. We still take the results obtained by BEM as independent references for the comparisons. In our finite difference simulation, the grid spacing is 0.04 km and the time step is 0.005 s. The number of grid points per shortest wavelength is about 10. Fig. 20 shows more detailed comparisons for three representative receivers indicated by triangles in Fig. 8. All comparisons shown in Figs 19 and 20 still show excellent agreement between the results by the two methods even for such an extreme topographic model.

Owing to the existence of extremely sloped edges at $x = 14$ and 16 km, strong and complex scattering of both body and Rayleigh waves by such sharp topographic changes can be observed very clearly on the synthetic seismograms shown in Fig. 19. Comparing the forward scattered waves, that is, those waves passed through the canyon, we find that the scattered body waves are much more prominent than those from Rayleigh wave, indicating that such sharp canyon can block the Rayleigh wave's propagation significantly, which is quite different from the case of smooth canyon discussed previously. It also can be seen that the largest amplitude of the ground motion is located at the source-side edge of the canyon and is caused by the superposition of the incoming Rayleigh wave and its scattering by the edge.

7.6 Results for a half-space with step shape topography

The last model with irregular topography considered is a half-space with step shape topography as illustrated in Fig. 9. This model is also an extreme topography model with a discontinuous topography. In this test model, there is a step falling down from 4 km to 0 km at

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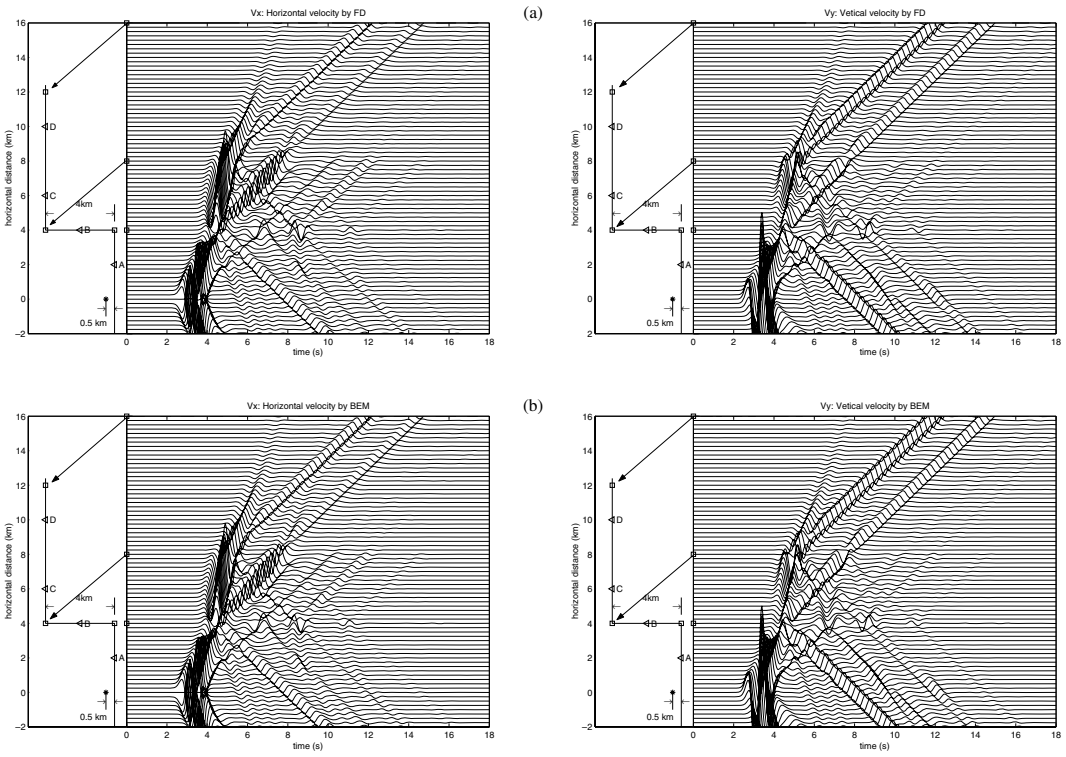


Figure 22. Trace records of the velocity components along the free surface of the step shape model. Left-hand side panel shows the horizontal velocity component, v_x; right-hand side panel shows the vertical velocity component, v_y. (a) Obtained by finite difference method, (b) Obtained by the boundary element method.

the free surface as shown in Fig. 9. The gridding scheme near the step corner is shown in Fig. 21.

In our finite difference simulation, the grid spacing is 0.05 km along the free surface and is about 0.045 km along η axis near the step corner, thus the number of grid points per shortest shear wavelength is about 10. The synthetic seismograms are calculated up to 20 s with a time step of $\Delta t = 0.005$ s.

Fig. 22 shows the comparisons of the synthetic seismograms calculated by our FDM implemented with the Traction Image method and the BEM, respectively. The detailed features of the waveform comparisons are shown in Fig. 23 for four selected receivers as indicated by triangle symbols in Fig. 9. All comparisons shown in these two figures display excellent agreements between the results by both methods even for such an extreme topography model, again further confirming the accuracy and wide-range applicability of our new method proposed in this study.

The more intuitive and visible displays of the wave's scattering process are the snapshots given in Fig. 24, where the whole process of wave propagation is clearly shown. From Figs 22–24, we can see that the scattering due to such a step in topography can generate strong Rayleigh waves.

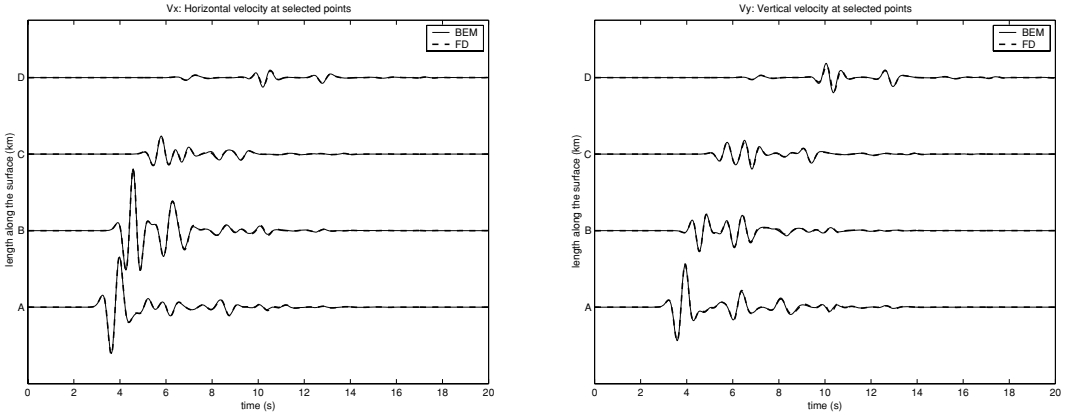


Figure 23. Detailed comparisons between the seismograms calculated with the finite difference (dashed line) and those from the boundary element method (solid line) of the step shape model at the selected observation points A (2, 4), B (4, 2), C (6, 0), and D (10, 0) as indicated in Fig. 9. The amplitudes are two times enlarged compared with Fig. (22).

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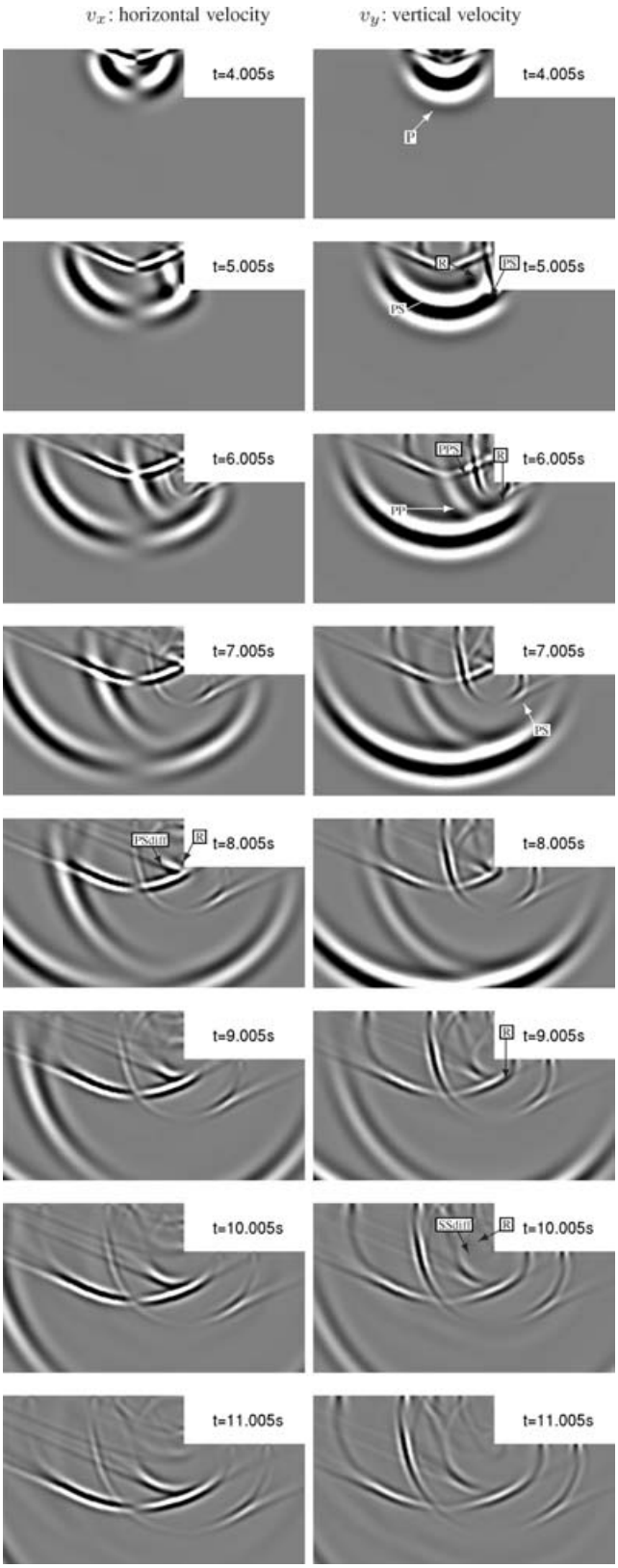


Figure 24. Snapshots showing wave propagation of the step shape model. The left-hand side panels are the horizontal velocity *Vx*, and the right-hand side panels are the vertical velocity *Vy*.

Table 1. Computational work to compute a derivative.

	Multiplies	Additions
4–2 MacCoramck	3	2
DRP/opt MacCoramck	5	4

8 DISCUSSION AND CONCLUSIONS

In this study, we have developed a new numerical method, named Traction Image method, to implement traction-free boundary conditions in finite difference simulation in the presence of surface topography.

To avoid the artificial scattering waves due to the staircase discretization of the irregular surface, we used the boundary-conforming grid to discretize the computational domain. Mathematically, such gridding is equivalent to a general curvilinear coordinate transformation. Since the finite difference scheme is implemented in curvilinear coordinates, a non-staggered finite difference scheme is needed, and so we used the DRP/opt MacCormack scheme.

To preserve the traction-free surface boundary conditions when calculating the derivatives of the stress components, we proposed the Traction Image method, in which the momentum conservation equations are recast into a conservative form and the traction components on the ghost grid points is set being antisymmetric about the free surface. In addition, the physical traction-free boundary conditions provide a constraint on the derivatives of the velocity components along the free surface, which leads to a solution to calculate the derivative of the velocity components by a compact scheme. With such traction-imaging technique, we can accurately account for traction-free boundary conditions in the simulation. These proposed effective measures were implemented in the DRP/opt MacCormack scheme and 4/4 Compact MarCormack scheme with 4–6 LDDRK time integration.

To validate the accuracy, we conducted numerical tests on six selected examples by comparing our simulation results with those obtained by other independent accurate simulation methods. Although some of the examples are quite extreme, all tested results showed excellent agreements between our simulation results and those from our reference solutions. The results confirm that our method can accurately simulate seismic waves in the heterogeneous media with arbitrarily shaped topography.

As final note, we discuss the computer performance of the method. Firstly, the traction-imaging settings and the compact scheme are performed only on the boundary grid points, so the impact on the overall computing performance from the numerical treatment for free surface boundary is negligible. Secondly, our method is based on the eqs (7a–e) in curvilinear coordinates, which cause a little more computer effort compared with those based on eqs (6a and b) in the Cartesian coordinate. Although twice as many terms are present, the number of the spatial derivatives in

Table 2. Number of arithmetic operations to assemble $AW_{,\xi} + BW_{,\eta}$.

	Eqs (6a and b) Cartesian	Eqs (7a–e) Curvilinear	Cost increase
4–2 MacCormack	54	98	81 per cent
DRP/opt MacCormack	86	138	60 per cent

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eqs (7a–e) has only increased from 8 to 10 compared with eqs eqs (6a and b) (Fornberg 1988). Thus higher-order finite difference schemes, in which the performance is dominated by the calculation of the spatial derivatives, with eqs (7a–e) does not double the computer effort than with eqs (6a and b). Table 1 shows the computational work required to compute a derivative using the 4-2 MacCormack and the DRP/opt MacCormack schemes, and Table 2 shows the operation counts required to assemble the right-hand side of eqs (7a–e) and eqs (6a and b) once on one grid point. It can be seen that the computer cost only increases about 60 per cent for the DRP/opt MacCormack scheme, but a little more for the 4-2 MacCormack scheme. If taking the time integration scheme into account, the overall computer cost increases about 45 per cent for the DRP/opt MacCormack scheme with the 4–6 LDDRK scheme with eqs (7a–e) than with eqs (6a and b). Finally, the Traction Image technique proposed in this study can be also incorporated into other effective finite difference schemes, thus there is room to improve the computational performance.

The current study can be easily extended to 3-D problem.

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APPENDIX A: NORMAL VECTOR ON THE FREE SURFACE

In curvilinear coordinates, the covariant base vectors are

$$\begin{aligned}\vec{e}_1 &= \frac{\partial x}{\partial \xi}\vec{i}_1 + \frac{\partial y}{\partial \xi}\vec{i}_2,\\ \vec{e}_2 &= \frac{\partial x}{\partial \eta}\vec{i}_1 + \frac{\partial y}{\partial \eta}\vec{i}_2,\end{aligned} \tag{A1}$$

where $\vec{e}_1$ is parallel with the surface and $\vec{e}_2$ is normal to the surface. The contravariant base vectors

$$\begin{aligned}\vec{e}^1 &= \nabla\xi = \partial_x\xi\,\vec{i}_1 + \partial_y\xi\,\vec{i}_2,\\ \vec{e}^2 &= \nabla\eta = \partial_x\eta\,\vec{i}_1 + \partial_y\eta\,\vec{i}_2,\end{aligned} \tag{A2}$$

are perpendicular to the covariant base vectors

$$\vec{e}^2 \perp \vec{e}_1 \qquad \vec{e}^1 \perp \vec{e}_2. \tag{A3}$$

So $\vec{e}^2$ is perpendicular to the free surface. Then the unit normal vector to the free surface is

$$\vec{n} = \frac{\vec{e}^2}{|\vec{e}^2|} = \frac{\nabla\eta}{|\nabla\eta|} = \frac{\eta_{,x}\vec{i}_1 + \eta_{,y}\vec{i}_2}{|\nabla\eta|}, \tag{A4}$$

where $|\nabla\eta| = \sqrt{\eta_x^2 + \eta_y^2}$

APPENDIX B: CONSERVATIVE FORM OF THE MOMENTUM CONSERVATION EQUATIONS

For 2-D inhomogeneous linear isotropic elastic media, the tensor form of the first-order partial differential velocity-stress equations consist of Momentum conservation equation:

$$\rho\frac{\partial v}{\partial t} = \nabla\cdot\sigma + f, \tag{B1}$$

and the Generalized Hook's Law:

$$\frac{\partial \sigma}{\partial t} = \lambda I(\nabla\cdot v) + \mu(\nabla v + v\nabla). \tag{B2}$$

In curvilinear coordinates, operator $\nabla\cdot\mathbf{A}$ can be expressed in two forms (Thompson *et al.* 1985)

Conservative form:

$$\nabla\cdot\mathbf{A} = \frac{1}{J}\sum_{i=1}^{2}[J\vec{e}^i\cdot\mathbf{A}]_{,\xi^i} \tag{B3}$$

Non-conservative form:

$$\nabla\cdot\mathbf{A} = \sum_{i=1}^{2}\vec{e}^i\cdot\mathbf{A}_{,\xi^i} \tag{B4}$$

where $\xi^i \in \{\xi, \eta\}$, J is the Jacobian of the transformation, and $\vec{e}^i$ are the contravariant base vectors in eq. (A2).

Substituting the conservative form of operator ∇ into the momentum conservation equation (eq. B1), we can derive the conservative form of the momentum equation in curvilinear coordinates as

$$\begin{aligned}\rho v_{x,t} &= \tfrac{1}{J}\{[J(\xi_{,x}\sigma_{xx} + \xi_{,y}\sigma_{xy})]_{,\xi}\\ &\quad + [J(\eta_{,x}\sigma_{xx} + \eta_{,y}\sigma_{xy})]_{,\eta}\},\\ \rho v_{y,t} &= \tfrac{1}{J}\{[J(\xi_{,x}\sigma_{xy} + \xi_{,y}\sigma_{yy})]_{,\xi}\\ &\quad + [J(\eta_{,x}\sigma_{xy} + \eta_{,y}\sigma_{yy})]_{,\eta}\}.\end{aligned} \tag{B5a}$$

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